

Further Integration Techniques



Integration by substitution

- 1 Evaluate $\int 2x(x^2 + 1)^4 dx$ using the substitution $u = x^2 + 1$.
 - 2 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$ using the substitution $u = \sin x$.
-

Integration by parts

- 3 Evaluate $\int x e^x dx$ using integration by parts.
 - 4 Evaluate $\int x \ln x dx$ using integration by parts.
-

Repeated integration by parts

- 5 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
-

Integration using trigonometric identities

- 6 Evaluate $\int \cos^2 x dx$ using the identity $\cos^2 x = \frac{1 + \cos 2x}{2}$.
-

Integration using partial fractions

- 7 Evaluate $\int \frac{1}{x^2 - 1} dx$ by first expressing the integrand in partial fractions.
-

Evaluating a definite integral

- 8 Evaluate $\int_1^e \frac{\ln x}{x} dx$, using the substitution $u = \ln x$.
 - 9 Evaluate $\int_0^{\pi} x \sin x dx$ using integration by parts.
-

Integration involving inverse trigonometric results

- 10 Evaluate $\int \frac{1}{1 + x^2} dx$, and use it to evaluate $\int_0^1 \frac{1}{1 + x^2} dx$.

Integration by substitution with trigonometric functions

- 11 Evaluate $\int \sin^3 x \, dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.
- 12 Evaluate $\int \tan x \, dx$, using the substitution $u = \cos x$.

Integration by parts with a logarithmic integrand

- 13 Evaluate $\int_1^e \ln x \, dx$ using integration by parts.

Integration using a Weierstrass-style substitution

- 14 Evaluate $\int \frac{1}{4\cos^2 x} \, dx$ using the substitution $t = \tan\left(\frac{x}{2}\right)$, given that this substitution transforms $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2}{1+t^2} dt$, leading to the simplified integrand $\frac{2}{9+t^2}$.

Integration by parts with a trigonometric-exponential product

- 15 Evaluate $\int e^x \cos x \, dx$, applying integration by parts twice and solving the resulting equation for the original integral.

Finding the area between two curves using integration

- 16 Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$ (the interval on which $\sin x \geq \cos x$).
- 17 Evaluate $\int x^2 \ln x \, dx$ using integration by parts.
- 18 A particle's velocity is given by $v(t) = t \cos t$ metres per second. Find the particle's displacement over the interval $0 \leq t \leq \pi$, using integration by parts.
- 19 This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int \frac{x}{x^2 + 4} \, dx$. (b) Use integration by parts to find $\int \arctan x \, dx$. (c) Hence, using the result of part (b) and the standard result $\int \frac{1}{1+x^2} \, dx = \arctan x + C$, evaluate $\int_0^1 \arctan x \, dx$, giving an exact answer in terms of π and $\ln$.

Reduction formulas

- 20 Given the reduction formula $I_n = \int x^n e^x dx = x^n e^x - n I_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x dx$.